

Quadratic functions and technology



1

a

i $(8, 7)$

ii $(0, 263)$

b

i $(-6, -9)$

ii $(0, 135)$

c

i $(-6, 4)$

ii $(0, -140)$

d

i $(2, -1)$

ii $(0, -13)$

e

i $\left(-\frac{1}{2}, \frac{33}{4}\right)$

ii $(0, 9)$

f

i $\left(-\frac{3}{8}, -\frac{25}{16}\right)$

ii $(0, -1)$

2

a

i $(0, 16)$

ii No

iii Increasing

b

i $(0, -12)$

ii No

iii Decreasing

c

i $(0, 125)$

ii Yes, 5, -5

iii Decreasing

d

i $(0, -64)$

ii Yes, 4, -4

iii Increasing

3

a No

b Yes

c Yes

d No

4 $c = 25$

5

a

i $-\frac{5}{2}$

ii $\frac{25}{4}$

iii $\left(-\frac{5}{2}, 0\right)$

b

i 1, 5

ii -5

iii $(3, 4)$

c

i 1, 7

ii -7

iii $(4, 9)$

6

a

i $(-1, 3)$

ii $(1, 3)$

b 2 units

7 Maximum value

8

a Minimum

b $(0, -4)$ 

9 $y = -\frac{81}{4}$

10

a -3.1 b -16.61

11

a

i $0, 6$ ii 0 iii 3 iv $(3, 9)$

b

i $0, -2$ ii 0 iii -1 iv $(-1, -1)$

c

i $0, -4$ ii 0 iii -2 iv $(-2, 4)$

d

i $-1, 3$ ii -3 iii 1 iv $(1, -4)$

e

i $-3, 1$ ii -3 iii -1 iv -4

f

i $2, 6$ ii -24 iii 4 iv 8

12

a 2.56 b 4.45

c

i Yes ii No iii No

d

 $5.12, 0.00$

13

a 2.75 b -9.10

c

i No ii Yes iii No

d

 $-0.42, 5.91$

14

a 1.55 b 2.95

c

i No ii Yes iii No iv No

d

 $3.58, -0.47$

15

a -1.38 b 0.84

c

i No ii No iii No iv Yes

d

 $-0.69, -2.08$

16 Move the graph downwards by 5 units.

17 Move the graph to the right by 3 units.



18 Narrows the graph

19

a $y = x^2 + 2$

b $y = -x^2$

c $y = -x^2 - 2$

d $y = (x + 2)^2 + 2$

20 $y = (x - 2)^2 + 3$

21

a $x = 0, x = \frac{8}{7}$

b $x = -\frac{1}{2}, x = 3$

c $x = \frac{3}{2}, x = -1$

d $x = -\frac{1}{4}, x = 5$

e $x = -1$

f $x = -4, x = -3$

22

a 1

b 2

c 0

23 2.000, 1.667

24 1.584, -0.859

Additional questions

25

a (0, 0)

b The initial distance fallen by the object

26

a 0, 30

b 15

c The largest rectangular area that can be fenced off with 60 m of wire.

27 ± 14 m/s

28

a 176.4 m

b 6 s

c 12 s

29



- a 900 m
- b Initially, the object has not fallen any distance.
- c 13.55 s
- d After 13.55 s the object is on the ground.

30

- a 4 s
- b 10 s